Foundational React Projects: Getting Started

01

Todo App

Focus: React + Local State

- Initialize with `npx create-react-app`
- Define functional components for Input and List
- Use useState to track an array of todo objects
- Implement handlers for adding, deleting, and toggling status

02

Counter App

Focus: Hooks & Events

- Master useState for managing numeric counts
- Implement onClick event handlers for increment/decrement
- Learn functional updates for state consistency
- Practice conditional rendering for limit alerts

03

Product Listing

Focus: UI & Components

- Create reusable Card components with props
- Map over static data arrays to render dynamic lists
- Style with Bootstrap grid for responsiveness
- Handle UI states like empty searches or loading

Roadmap: Full-Stack Project Tiers

Beginner

Todo App
React + Local State

Counter App
State management with Hooks

Static Product Listing
UI Design & Components

Intermediate

Student Mgmt System
Full CRUD Operations

Blog App
Authentication + Posting

REST API
Express + MongoDB Integration

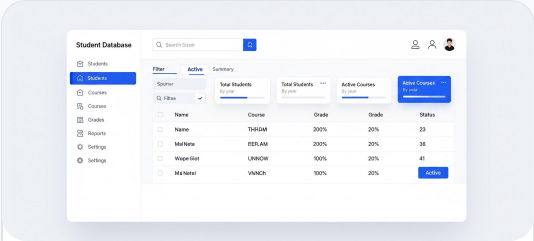
Advanced

E-commerce App
Cart logic + Payment flow

Social Media App
Feed, Likes & Real-time updates

Admin Dashboard
Complex Analytics & Charts

Roadmap: Intermediate Full-Stack Implementation



01 Student Mgmt System

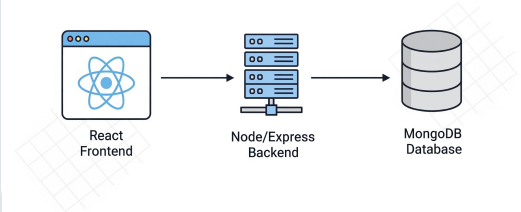
CRUD OPERATIONS

- Define Mongoose Schema for Student (Name, Age, Course)
- Build API endpoints for Create, Read, Update, Delete
- Frontend: Axios calls for form submission & list rendering
- Add logic for data persistence in MongoDB

02 Blog Application

AUTH + POSTING

- Implement JWT-based Login & Registration
- Create Secure Routes for Posting content
- Handle User Sessions & Cookies
- Dynamic rendering of user-specific post feeds

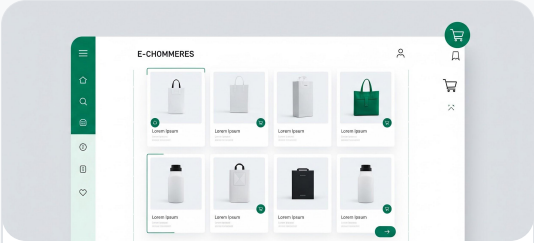


03 REST API Design

EXPRESS + MONGO INTEGRATION

- Establish Express server with CORS middleware
- Design RESTful resource paths (GET, POST, PUT, DELETE)
- Database Connection via Mongoose ODM
- Environment variable config for DB credentials

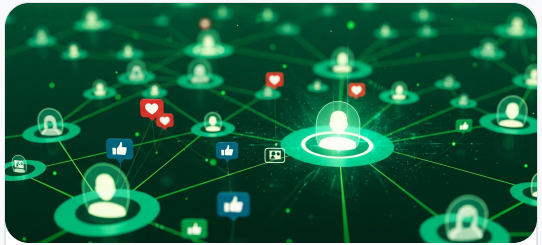
Roadmap: Advanced Full-Stack Architecture



01 E-commerce App

CART LOGIC + PAYMENTS

- Develop persistent shopping cart state management
- Integrate Stripe API for secure checkout flows
- Handle inventory updates and order confirmation emails
- Build user order history and tracking dashboard



02 Social Media App

REAL-TIME ENGAGEMENT

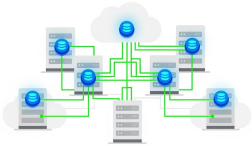
- Implement WebSocket (Socket.io) for live feed updates
- Design "Like" and "Comment" systems with optimistic UI
- Build image upload pipeline with cloud storage (S3)
- Algorithm for personalized user discovery and follow logic

03 Admin Dashboard

ANALYTICS & COMPLEX DATA

- Interactive data visualization with Chart.js or D3.js
- Aggregated reporting using MongoDB pipelines
- Role-based access control (RBAC) for administration
- Log monitoring and system health performance metrics

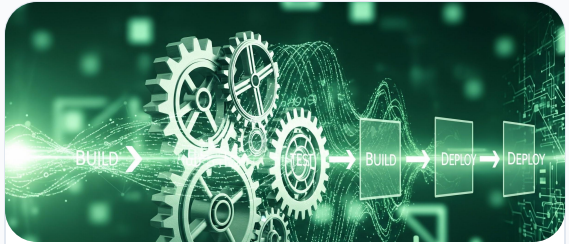
Final Stage: Production & Deployment



Cloud Hosting

Infrastructure

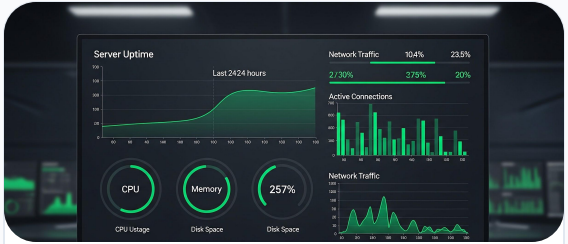
- Deploy Frontend to Netlify/Vercel
- Host Node.js on Render or AWS
- Configure Production MongoDB Atlas
- Setup SSL/HTTPS certificates



CI/CD Pipeline

Automation

- Automate testing (GitHub Actions)
- Scripted build & deploy flows
- Env variable management
- Version control tagging (v1.0.0)



Monitoring

Maintenance

- Real-time error logging w/ Sentry
- Performance & uptime tracking
- User analytics implementation
- Periodic database backups